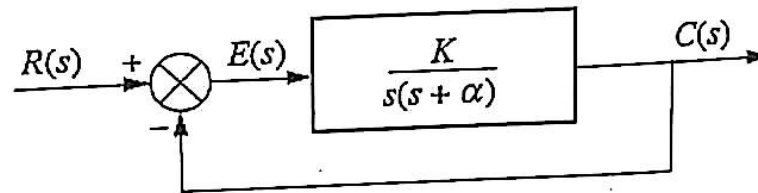


CLO # 2: Analyze the transient and steady state response of a feedback system.

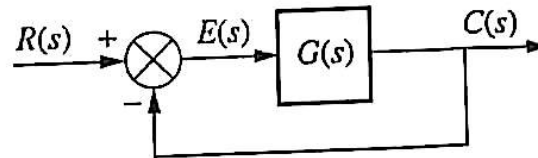
Q1: For the system shown below, design K and α to yield a settling time of 0.4 second and a 30% overshoot. [10 marks]



CLO # 3: Analyze the stability of a system

Q2: For the unity feedback system shown below,

[10 marks]



$$G(s) = \frac{K(s+6)}{s(s+1)(s+4)}$$

- Find the system's closed loop transfer function, $T(s) = C(s) / R(s)$.
- Hence, find the range of K for stability.

$$\omega_n = \sqrt{b}$$

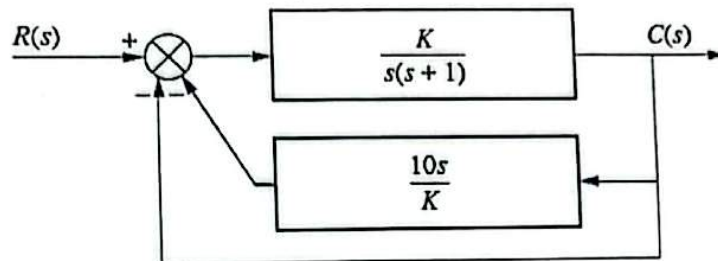
$$\zeta = \frac{a}{2\sqrt{b}}$$

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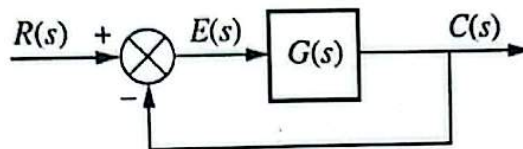
CLO # 2: Analyze the transient and steady state response of a feedback system.

Q3: Given the system shown below:

[10 marks]



a. Use block diagram reduction to express it in the following form and find $G(s)$.



- b. Hence, use $G(s)$ to find velocity constant and design the value of K so that for a unit-ramp input of $tu(t)$, there will be a constant steady state error of 0.15.
- c. Evaluate the steady-state error of the following system caused by the unit-step disturbance $D(s)$ only.

$$\left(\text{Hint: } e(\infty) = \lim_{s \rightarrow 0} \left[\frac{s}{1 + G_1(s)G_2(s)} \right] R(s) - \lim_{s \rightarrow 0} \left[\frac{sG_2(s)}{1 + G_1(s)G_2(s)} \right] D(s) \right)$$

